

LESSON 20 / 第二十課

Urban Mycorrhizae: From Host Communities to the Cortex Exchange

城市菌根：從寄主群落到皮層交換

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Most urban woody taxa form arbuscular mycorrhizae; selected families form ectomycorrhizae. The association is a carbon-for-nutrient exchange inside the root cortex, not a product poured from a bag. High available phosphorus suppresses AM colonisation by reducing the host’s strigolactone signal and carbon bid. Maltz and Treseder found inoculant success contingent, not guaranteed. Field assay remains staining plus grid-line intersect counts. Ranked interventions start with volume, oxygen, density and P status; inoculant is last.

多數城市木本分類群形成叢枝菌根；選定科形成外生菌根。該共生是根皮層內的碳換養分，不是從袋中倒出的產品。高有效磷透過降低宿主的獨腳金內酯訊號與碳出價抑制叢枝定殖。Maltz 與 Treseder 發現接種成功是有條件的，不是保證的。田間測定仍是染色加格線相交計數。有排序的干預始於體積、氧、密度與磷狀態；接種劑放在最後。

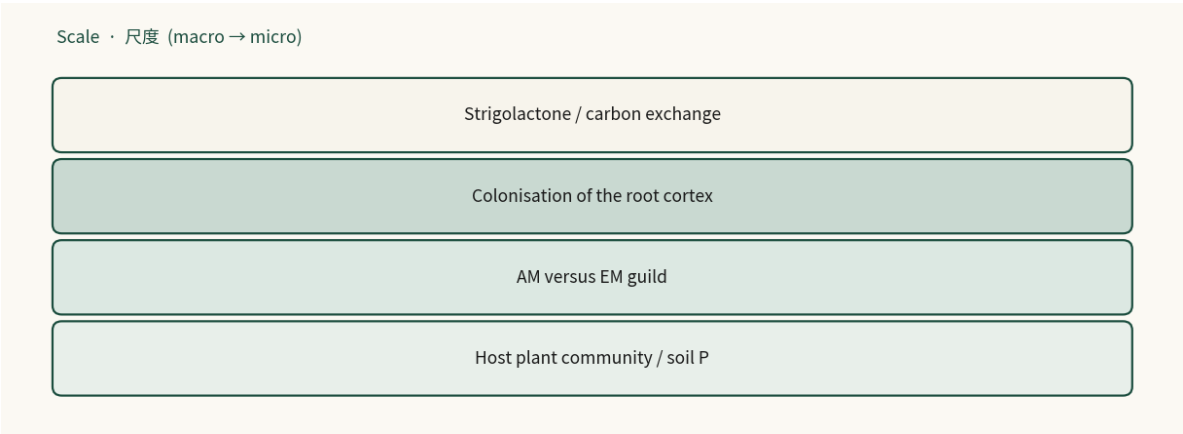


Figure 1. Nested scales from the host community to the cortex exchange.

圖 1. 由寄主群落到皮層交換的嵌套尺度。

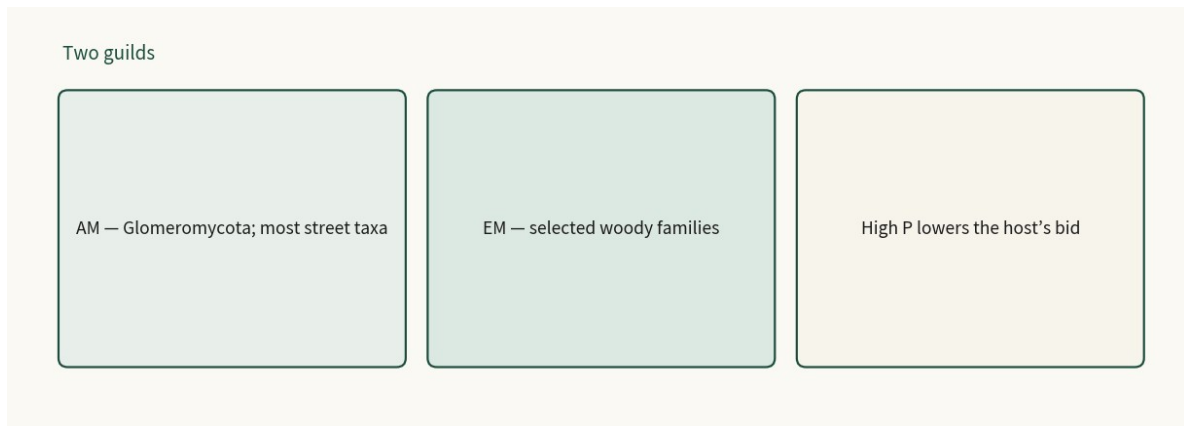


Figure 2. AM, EM and phosphorus suppression.

圖2. 叢枝、外生與磷抑制。

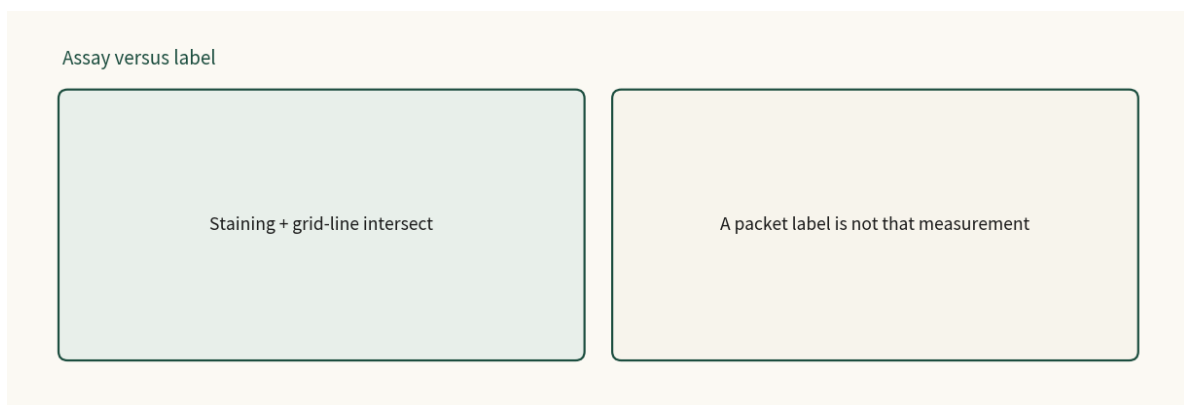


Figure 3. Field assay versus product label.

圖3. 田間測定對產品標籤。

1. Scope

1. 範圍

The object is published mycorrhizal ecology of woody plants as it applies to urban soils. A commercial inoculant is an intervention inside that object, not the object.

對象是適用於城市土壤的木本植物已發表菌根生態。商業接種劑是該對象之內的干預，不是對象本身。

2. Guilds and P suppression

2. 功能群與磷抑制

Arbuscular mycorrhizae dominate most urban street taxa; ectomycorrhizae dominate selected families (Fagaceae, Pinaceae, some Myrtaceae) [1]. High available P reduces AM colonisation via lower strigolactone exudation and a smaller carbon allocation from the host, not via contact poison [1,2]. Compacted, hypoxic and saline soils further narrow the partner set [3].

叢枝菌根主導多數城市行道分類群；外生菌根主導選定科（殼斗科、松科、部分桃金娘科）[1]。高有效磷透過較低獨腳金內酯分泌與較小宿主碳分配降低叢枝定殖，不是靠接觸毒殺 [1,2]。壓實、缺氧與鹽土進一步收窄夥伴集合 [3]。

3. Inoculant and assay

3. 接種劑與測定

Maltz and Treseder's synthesis found restoration inoculants contingent on site and taxa, not universally effective [4]. Field assay is staining and grid-line intersect, not a packet label [5]. Ranked interventions: recover volume, oxygen and density; correct P and salt; then argue about inoculant.

Maltz 與 Treseder 的綜述發現復育接種視立地與分類群而定，不是普遍有效 [4]。田間測定是染色與格線相交，不是包裝標籤 [5]。有排序的干預：先恢復體積、氧與密度；校正磷與鹽；然後才爭論接種劑。

4. Limits

4. 限度

Urban AM community composition data remain thinner than agricultural AM data. Matching an inoculant isolate to a street taxon in a paved pit is still more hope than measurement.

城市叢枝群落組成數據仍薄於農業叢枝數據。在鋪面樹穴裡把接種分離株配到行道分類群，仍是希望多於測量。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. Most urban street taxa form:

1. 多數城市行道分類群形成：

- A. Only EM.
- A. 只有外生。
- B. AM.
- B. 叢枝。
- C. No mycorrhizae by definition.
- C. 定義上無菌根。
- D. Only lichens.
- D. 只有地衣。

2. High available P suppresses AM primarily by:

2. 高有效磷抑制叢枝，主要透過：

- A. Contact poison.
- A. 接觸毒殺。
- B. Reducing the host's signal and carbon bid.
- B. 降低宿主訊號與碳出價。
- C. Raising t/R.
- C. 提高 t/R。
- D. Increasing flash density.
- D. 增加閃擊密度。

3. Maltz and Treseder found inoculants:

3. Maltz 與 Treseder 發現接種劑：

- A. Universally successful.
- A. 普遍成功。
- B. Contingent, not guaranteed.
- B. 有條件，不保證。
- C. Illegal.
- C. 非法。
- D. Equal to P88.
- D. 等於 P88。

4. Field assay uses:

4. 田間測定使用：

- A. Packet labels.
- A. 包裝標籤。
- B. Staining and grid-line intersect counts.
- B. 染色與格線相交計數。
- C. Only i-Tree.
- C. 只有 i-Tree。
- D. t/R.
- D. t/R。

5. Ranked intervention starts with:

5. 有排序的干預始於：

- A. Inoculant packets.
- A. 接種劑包裝。
- B. Volume, oxygen, density and P status.
- B. 體積、氧、密度與磷狀態。
- C. Paint.
- C. 漆。
- D. Lightning conductors.
- D. 避雷導體。

6. EM is typical of:

6. 外生典型於：

- A. All street taxa.
- A. 所有行道分類群。
- B. Selected woody families.
- B. 選定木本科。
- C. Palms only.
- C. 只有棕櫚。
- D. All herbs.
- D. 所有草本。

7. A label is:

7. 標籤是：

- A. A colonisation measurement.
- A. 定殖測量。

B. Not a colonisation measurement.

B. 不是定殖測量。

C. Wall 2.

C. 第 2 壁。

D. A flare datum.

D. 幹基基準。

8. Hypoxic compacted soil:

8. 缺氧壓實土：

A. Widens the partner set.

A. 擴大夥伴集合。

B. Narrows the partner set.

B. 收窄夥伴集合。

C. Equals 3-30-300.

C. 等於 3-30-300。

D. Equals QTRA.

D. 等於 QTRA。

9. The exchange is located:

9. 交換位於：

A. In the catalogue.

A. 在目錄裡。

B. Inside the root cortex (and EM mantle / Hartig net).

B. 在根皮層內（以及外生菌套／哈蒂網）。

C. In the lightning path.

C. 在雷電路徑。

D. In the tomogram colour.

D. 在斷層顏色。

10. Matching an isolate to a paved-pit street taxon is:

10. 把分離株配到鋪面樹穴行道分類群是：

A. Usually a completed measurement.

A. 通常是已完成的測量。

B. Still more hope than measurement.

B. 仍是希望多於測量。

C. Required by Choat 2012.

C. Choat 2012 所要求。

D. A cable termination.

D. 鋼纜錨頭。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

[1] Smith S.E., Read D.J. (2008). Mycorrhizal Symbiosis. 3rd ed.

[2] Carbonnel S., Gutjahr C. strigolactone and P-control of AM.

[3] Day S.D. et al. (2010). urban rhizosphere constraints.

[4] Maltz M.R., Treseder K.K. (2015). inoculant efficacy in restoration.

[5] McGonigle T.P. et al. grid-line intersect method.

[6] Bainard L.D. et al. urban mycorrhizal community studies.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。